

Computer vision exam program

MIPT, spring 2024

1. Image formation
 - a. Human color perception
 - b. Reflected light amplitude formula
 - c. Color constancy
 - d. Metamerism
2. Color spaces: structure and applications
 - a. RGB
 - b. HSV
 - c. CMYK
3. Camera models
 - a. Color filter arrays (Bayer filter)
 - b. Camera shutter
4. Computer vision libraries
 - a. For images
 - b. For videos
 - c. For other purposes
5. Image histogram
 - a. Histogram equalization
 - b. Gamma correction
6. Morphology
 - a. Erosion, dilation, opening, closing
7. Keypoints
 - a. SIFT
8. RANSAC
9. Background subtractors
 - a. MOG
10. Convolution implementation
 - a. Hyperparameters
 - b. Image strides
 - c. Convolution matrix
11. Datasets for classification, detection and segmentation
 - a. ImageNet
 - b. COCO
12. Image classification
 - a. Architectures from AlexNet to SENet
 - b. MobileNet v1, v2
13. Detection
 - a. Statement, metrics
 - b. R-CNN

- c. YOLO
- 14. Segmentation
 - a. Statement, metrics
 - b. Architectures from FCN to UNet
 - c. Loss functions taxonomy
- 15. Attention
 - a. simple attention (any other than seq2seq or self-attention example)
 - b. self-attention
- 16. Vision Transformer (ViT)
 - a. Swin
- 17. Generative models
 - a. VAE
 - b. GAN
- 18. Video processing with NNs
 - a. Early and late fusion